Teardown Benchmark Study

Cabin sub-system - sealing, glazing and interior noise
Subject vehicle against a European long-haul benchmark

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Benchmark class	European long-haul tractor, comparable cab volume
Test condition	80 km/h constant speed, smooth asphalt, laden
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Summary

The cabin sealing and glazing package of a European long-haul benchmark was stripped and compared against the subject vehicle, with interior noise measured on both under identical conditions. The benchmark records approximately 63 dB(A) against 68 dB(A) on the subject vehicle at 80 km/h. The gap is attributed to the air-suspended cab mount and acoustic laminated glazing rather than to added absorptive mass.

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1 Scope and test method

This study covers the cab sealing system, glazing, cab mounting and interior absorptive treatment. It excludes the seat, the instrument panel and the HVAC system, which are reported separately. Both vehicles were measured before teardown and then stripped, so that measured noise can be attributed to components actually present.

1.1 Acoustic measurement

Interior sound pressure level was measured at the driver's ear position with a half-inch free-field microphone, at 80 km/h constant speed on smooth asphalt, laden to 80 per cent of GVW, with HVAC off and windows closed. Three runs in each direction were averaged. Octave band levels were recorded from 125 Hz to 8 kHz.

1.2 What a 5 dB(A) difference means

A 5 dB(A) difference at the driver's ear is approximately a 40 per cent reduction in perceived loudness. It is well above the threshold at which drivers report a difference in a paired comparison, and it is the kind of gap that appears in fleet purchase evaluations.

Method note. Noise was measured before teardown on both vehicles, so the levels reported in section 2.1 correspond to the complete as-built sealing and glazing package, not to a partially stripped cab.

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1.3 Sealing system comparison

Element	Subject vehicle	Benchmark D
Door seal	Single bulb, EPDM	Twin bulb, EPDM, separate aperture seal
Door seal count	1 per aperture	2 per aperture
Window glazing	Toughened, 4 mm	Laminated acoustic, 4.8 mm
Windscreen	Laminated, 5.5 mm	Laminated acoustic, 5.5 mm
Cab mount	Rubber, 4 point	Air-suspended, 4 point
Floor absorptive mass	4.8 kg	3.2 kg
Roof absorptive pad	1.9 kg	1.4 kg
Firewall treatment	Bitumen pad, 3.1 kg	Moulded barrier, 2.6 kg

The benchmark carries 4.6 kg LESS absorptive treatment than the subject vehicle while achieving a lower interior level. This is the central observation of the study: the gap is not closed by adding mass.

Finding 1.1. The benchmark achieves a quieter cab with less absorptive mass. Treatment added downstream of a path that has not been sealed returns very little, which is the likely explanation for the subject vehicle's heavier and less effective package.

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2 Cabin sub-system

The cabin sub-system comparison covers four areas: the door sealing system, the glazing specification, the cab mounting arrangement and the interior absorptive treatment. The first two are reported here; the cab mount is discussed in section 3 because its cost consequence dominates its acoustic contribution.

2.0.1 Door sealing

The benchmark fits two seals per door aperture: a primary bulb seal on the door and a secondary aperture seal on the body. The subject vehicle fits one. Measured leak rate under a 50 Pa pressure differential was 14.2 litres per second on the subject vehicle against 4.8 on the benchmark.

Measurement	Subject vehicle	Benchmark D
Seals per aperture	1	2
Leak rate at 50 Pa (l/s)	14.2	4.8
Door closing effort (N)	62	71
Seal mass per aperture (kg)	0.84	1.46

2.0.2 Glazing

The benchmark uses acoustic laminated glass in the door windows, with a viscoelastic interlayer tuned to the coincidence frequency range of the pane. The subject vehicle uses toughened glass. The difference is 0.9 kg per window and an estimated INR 1,840 per vehicle across both doors.

Finding 2.1. Door-seal and acoustic-glass changes are the two elements of the benchmark's cabin NVH package. Measured results are reported in section 2.1.

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2.1 NVH bench results

Interior sound pressure level was measured on both vehicles under identical conditions. The benchmark cabin recorded approximately 63 dB(A) at the driver's ear at 80 km/h constant speed, against 68 dB(A) on the subject vehicle - a gap of 5 dB(A), attributed to the air-suspended cab mount and the acoustic laminated glazing.

Condition	Subject vehicle	Benchmark D	Difference
80 km/h constant, laden	68.0 dB(A)	63.0 dB(A)	-5.0
60 km/h constant, laden	64.2 dB(A)	60.1 dB(A)	-4.1
Idle, in cab	51.4 dB(A)	49.8 dB(A)	-1.6
Full load acceleration, peak	74.6 dB(A)	70.2 dB(A)	-4.4

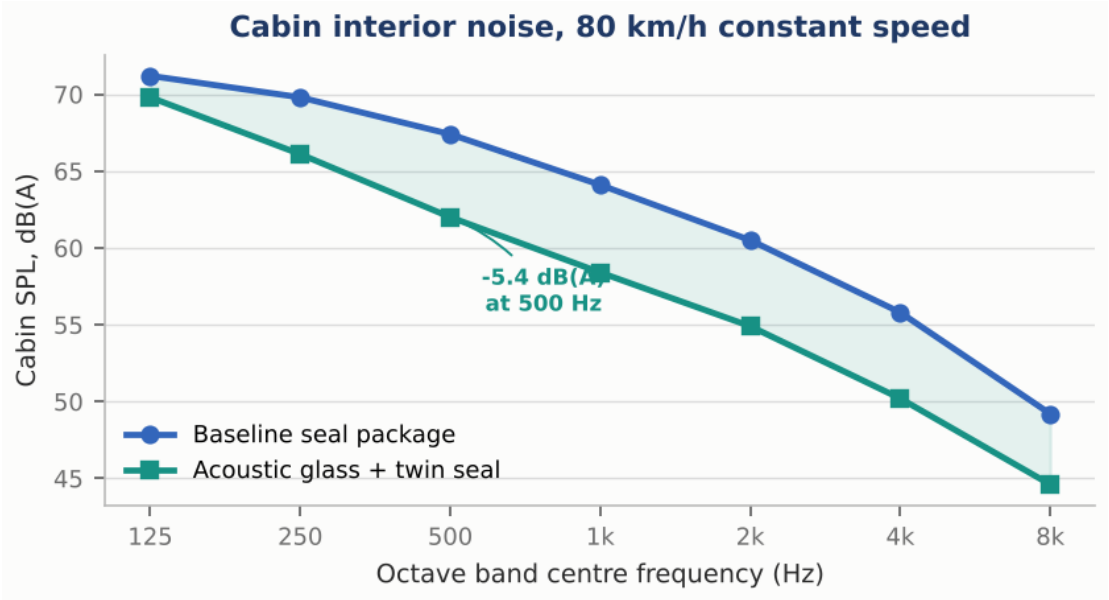


Figure 1 - Cabin interior noise by octave band at 80 km/h. The largest difference is in the 500 Hz to 2 kHz range, where door sealing and glazing dominate the transmission path.

Finding 2.2. The gap is concentrated between 500 Hz and 2 kHz, which is the range controlled by sealing and glazing rather than by structural transmission. This supports a sealing and glazing intervention over added absorptive mass.

3 Cost and mass

Element	Subject (INR)	Benchmark D (INR)	Mass delta (kg)
Door seals	1,240	2,180	+1.24
Door glazing	2,460	4,300	+1.80
Windscreen	6,840	7,920	+0.40
Cab mount	8,420	26,400	+6.80
Absorptive treatment	3,180	2,640	-4.60
Total	22,140	43,440	+5.64

The air-suspended cab mount accounts for 84 per cent of the cost difference and contributes an estimated 1.4 dB(A) of the 5 dB(A) gap. The seal and glazing changes together account for an estimated 3.1 dB(A) at 16 per cent of the cost difference.

Finding 3.1. Seal and glazing changes deliver roughly three fifths of the acoustic benefit for roughly one sixth of the cost. The cab mount is not cost-effective as an NVH measure on this evidence.

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4 Opportunities

Ref	Opportunity	Est. benefit	Cost (INR)	Class
CB-11-01	Twin bulb door seal with aperture seal	-1.8 dB(A)	+940	Running change
CB-11-02	Acoustic laminated door glazing	-1.3 dB(A)	+1,840	Running change
CB-11-03	Reduce floor absorptive mass once sealed	-1.6 kg	-420	Running change
CB-11-04	Moulded firewall barrier in place of bitumen	-0.5 kg	-180	Running change
CB-11-05	Air-suspended cab mount	-1.4 dB(A)	+17,980	Not recommended

Finding 4.1. CB-11-01 to CB-11-04 together give an estimated 3.1 dB(A) at a net cost of INR 2,180 and a mass reduction of 1.56 kg, because the absorptive mass reductions are enabled by the sealing improvement and partly offset its cost.

4.1 Dependency

CB-11-03 and CB-11-04 must not be taken without CB-11-01. Reducing absorptive mass on an unsealed cab would worsen interior noise. Submitted individually, the two mass reductions would be assessed as degrading NVH and closed.

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5 Conclusions

- The benchmark cabin is 5 dB(A) quieter at 80 km/h while carrying 4.6 kg less absorptive treatment.
- The gap sits between 500 Hz and 2 kHz, the range controlled by sealing and glazing rather than structural transmission.
- Seal and glazing changes deliver about three fifths of the benefit for about one sixth of the cost difference.
- The air-suspended cab mount dominates cost and is not recommended as an NVH measure.
- Four opportunities form a dependent package: the mass reductions are only valid once the sealing change is made.

Recommendation. Submit CB-11-01 to CB-11-04 as a single package. Record the dependency so that the two mass reductions are not assessed in isolation.

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